

Functional Specification

Serverless HR Lookup Application

Student: Nitish Kumar AWS Region: us-east-1

1 Purpose and Scope

This application provides an authenticated employee lookup portal. A signed-in user enters an Employee ID and receives the matching record from DynamoDB through a protected API. The browser never accesses DynamoDB directly.

2 Architecture

The browser calls API Gateway. GET / invokes hr-ui-function and returns the application page. GET /employee/{id} is protected by SecureHRAuthorizer. After Cognito validates the ID token, API Gateway invokes hr-backend-lookup, which reads the Employees DynamoDB table using its IAM execution role.

Architecture diagram: [architecture-diagram.svg](#)

3 Authentication

The SPA uses Cognito Managed Login with OAuth 2.0 Authorization Code Grant and PKCE. The browser creates a verifier and challenge, receives an authorization code, exchanges the code and verifier for tokens, and sends the Cognito ID token in the Authorization header. Authentication diagram: [authentication-flow.svg](#)

4 AWS Configuration

Resource	Configuration
API Gateway	serverless-hr-api, stage prod
API URL	https://ioy99699v8.execute-api.us-east-1.amazonaws.com/prod
UI Lambda	hr-ui-function
Backend Lambda	hr-backend-lookup
DynamoDB	Employees; partition key EmployeeID (String)
IAM role	HRBackendLambdaRole; dynamodb:GetItem only
Cognito User Pool	us-east-1_yLlaBkBX6
App Client	6v46kadj8g17udr54i6j2a50vb; public, no secret
Managed Login	https://us-east-1yllabkxbx6.auth.us-east-1.amazoncognito.com
User sub	24b88468-2061-7078-20b0-97ee47d4926c

5 Data Model

Attribute	Type	Purpose
EmployeeID	String	Partition key and lookup identifier
Name	String	Employee name
Salary	Number	Employee salary
DateOfJoin	String	Join date
Description	String	Description or student AWS ARN

6 Testing Evidence

Test	Result
Application access	UI opens over HTTPS
Cognito authentication	Managed Login and PKCE work
Valid lookup 1001	Alice Chan record displayed
Student lookup 1005	Nitish Kumar and IAM user ARN displayed
Invalid ID 99999	Employee not found displayed
Unauthorized API access	HTTP 401 Unauthorized returned

7 Submission Files

GitHub repository: <https://github.com/nitishhrms/serverless-hr-lookup>

API export: api-export.json

Architecture diagram: architecture-diagram.svg

Authentication flow: authentication-flow.svg

AI assistance record: docs/ai-tools.md

Screenshot evidence: organized in the final submission folder.

8 Cleanup

After all screenshots and submission artifacts are captured, remove or disable assignment-only API Gateway, Lambda, DynamoDB, Cognito, IAM, and EC2 resources as directed by the instructor.